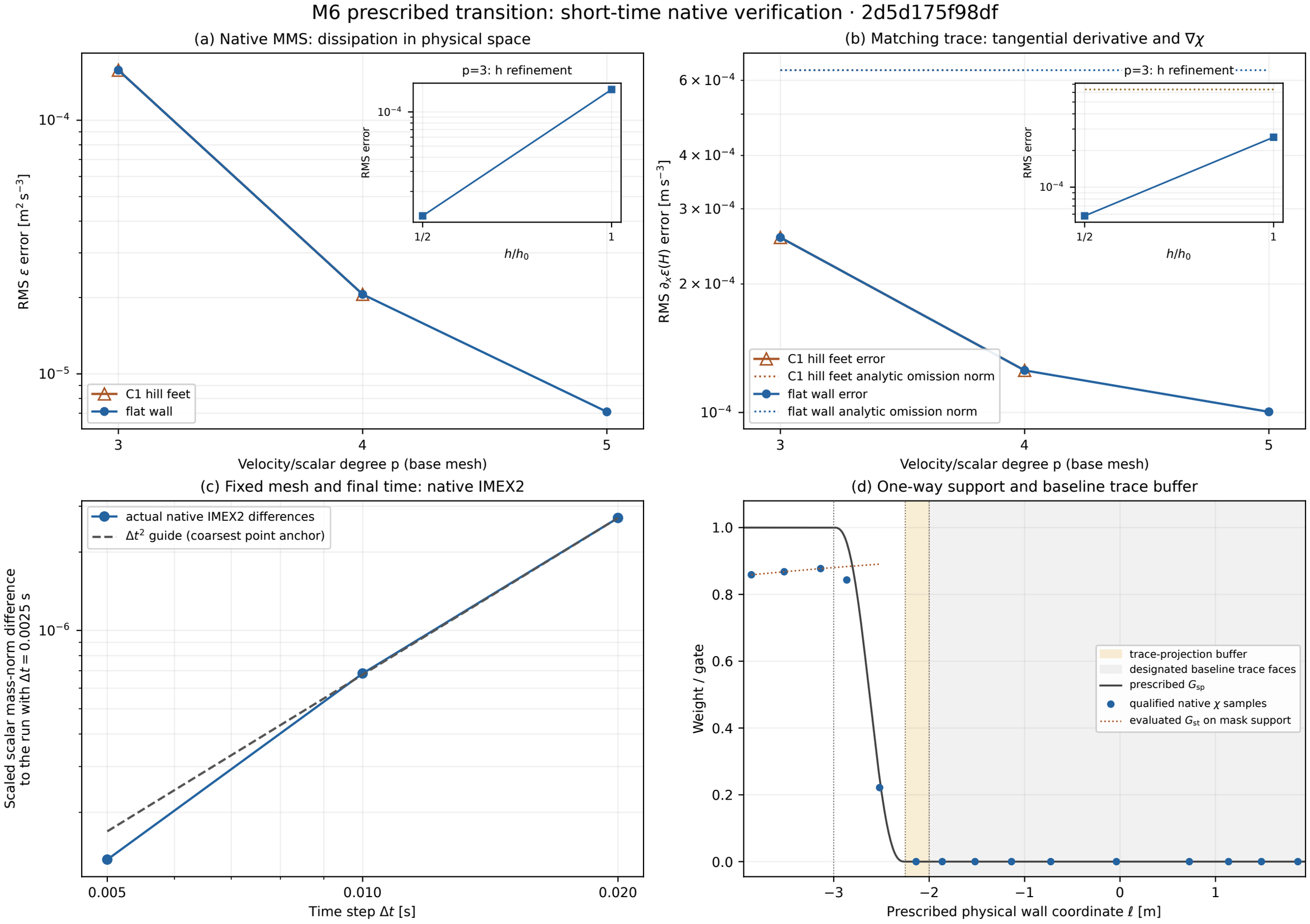
M6 prescribed transition: short-time native verification · 2d5d175f98df
(a) Native MMS: dissipation in physical space
RMS ε error [m² s⁻³]
Velocity/scalar degree p (base mesh)
C1 hill feet
flat wall
p=3: h refinement
RMS error
h/h₀
(b) Matching trace: tangential derivative and ∇χ
RMS ∂ₓε(H) error [m s⁻³]
C1 hill feet error
C1 hill feet analytic omission norm
flat wall error
flat wall analytic omission norm
(c) Fixed mesh and final time: native IMEX2
Scaled scalar mass-norm difference to the run with Δt = 0.0025 s
actual native IMEX2 differences
Δt² guide (coarsest point anchor)
Time step Δt [s]
(d) One-way support and baseline trace buffer
Weight / gate
trace-projection buffer
designated baseline trace faces
prescribed G_sp
qualified native χ samples
evaluated G_st on mask support
Prescribed physical wall coordinate ℓ [m]